STUDY BUDDY STEM AI: AN AI-POWERED PERSONALIZED, ADAPTIVE AND MULTIMODAL STEM LEARNING ECOSYSTEM FOR SRI LANKAN STUDENTS

INDIVIDUAL THESIS

Handwritten Content Analysis and Knowledge Synthesis

Dilshan WA

IT22292568

Group ID R26-IT-009

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Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

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DECLARATION AND SUPERVISOR CERTIFICATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Student ID: IT22292568

Signature: _____ Date: _____

The above candidate has carried out research for the bachelor's degree dissertation under my supervision.

Supervisor: _____

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Co-supervisor: _____

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ABSTRACT

Handwritten science notes can contain recognition ambiguities, incomplete explanations and misconceptions that remain unnoticed when the notes are converted directly into revision resources. This thesis presents the Handwritten Content Analysis and Knowledge Synthesis component of Study Buddy STEM AI for English-language Grade 10 and Grade 11 science learning in Sri Lanka. The component uses Gemma 4 Cloud to transcribe note images and support revision-resource generation, while a local Llama 3.2 3B model uses retrieved curriculum evidence to assess claims and prepare corrected notes. The workflow separates factual claims, combines semantic and lexical retrieval, requires valid evidence references for educational verdicts, and distinguishes incorrect content from insufficient evidence and processing failure. Conservative reconstruction retains supported content; a repair pathway subjects proposed corrections to fresh verification before acceptance. The accepted note becomes the source for flashcards, mind maps, structured notes and related multimedia learning. The research design specifies teacher annotation, stage-specific measures, paired baselines and end-to-end evaluation, including checks for unsupported additions and useful-content retention. Component-level empirical evaluation remains pending, so no accuracy, model-ranking or learning-improvement result is claimed. The present contribution is an integrated, inspectable architecture and a reproducible evaluation plan for curriculum-bounded note analysis. Future validation must determine whether its evidence controls improve reliability enough to justify their additional computational and review costs.
Keywords: handwritten note analysis; retrieval-augmented generation; claim verification; knowledge synthesis; science education.

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I acknowledge the contribution of the Study Buddy STEM AI group to the wider research project and the shared technical context within which this individual component is developed. This thesis concentrates on my work on handwritten content analysis and knowledge synthesis, while recognising the separate responsibilities of the other component authors.

I am grateful to the researchers and open-source communities whose published methods, model documentation and software make this investigation possible. Their contributions are acknowledged through the references and through the distinction between established methods and the proposed individual contribution.

I also acknowledge the assistance of generative AI in preparing and formatting this document, as described in Appendix F. The responsibility for reviewing the research account, completing the evaluation and submitting an accurate thesis remains with me.

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LIST OF ABBREVIATIONS

Abbreviations used in the thesis

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
BM25	Best Matching 25
CER	Character Error Rate
CPU	Central Processing Unit
FAISS	Facebook AI Similarity Search
FEVER	Fact Extraction and VERification
GCE O/L	General Certificate of Education Ordinary Level
GPU	Graphics Processing Unit
ID	Identifier
JSON	JavaScript Object Notation
LLM	Large Language Model
MRR	Mean Reciprocal Rank
NLP	Natural Language Processing
OCR	Optical Character Recognition
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
RAM	Random Access Memory
RRF	Reciprocal Rank Fusion
SDG	Sustainable Development Goal
URL	Uniform Resource Locator
WER	Word Error Rate

1 INTRODUCTION

1.1 Overall Project Context

Study Buddy STEM AI is a group research project that brings together four forms of learning support for Sri Lankan students preparing for the General Certificate of Education Ordinary Level examination. These are curriculum-grounded storytelling, conversational tutoring, emotion-aware adaptive assessment, and handwritten content analysis and knowledge synthesis. The shared purpose is to connect learning activities that students would otherwise carry out using separate applications. A learner can explore an idea, ask a question, practise its application, and return to personal notes for revision. The group manuscript describes this wider system and provides the context for the individual component examined here [1].

This thesis concerns the fourth component only. Its starting point is a student's own English science note, rather than a question created by the tutor. A note can contain a useful explanation alongside an incorrect statement, an omitted condition, or an ambiguous symbol. Converting that note into polished study material without inspecting its content could preserve the original misconception. The research therefore places evidence-based checking between text capture and resource generation. The intended users are Grade 10 and Grade 11 learners studying science in English; teachers provide the expert judgement required for subsequent validation.

Figure 1.1 locates the individual component within the group project. The surrounding components are integration context and are not claimed as the individual author's work. Shared access to curriculum resources does not imply that every component uses an identical model, database configuration, or evaluation procedure.

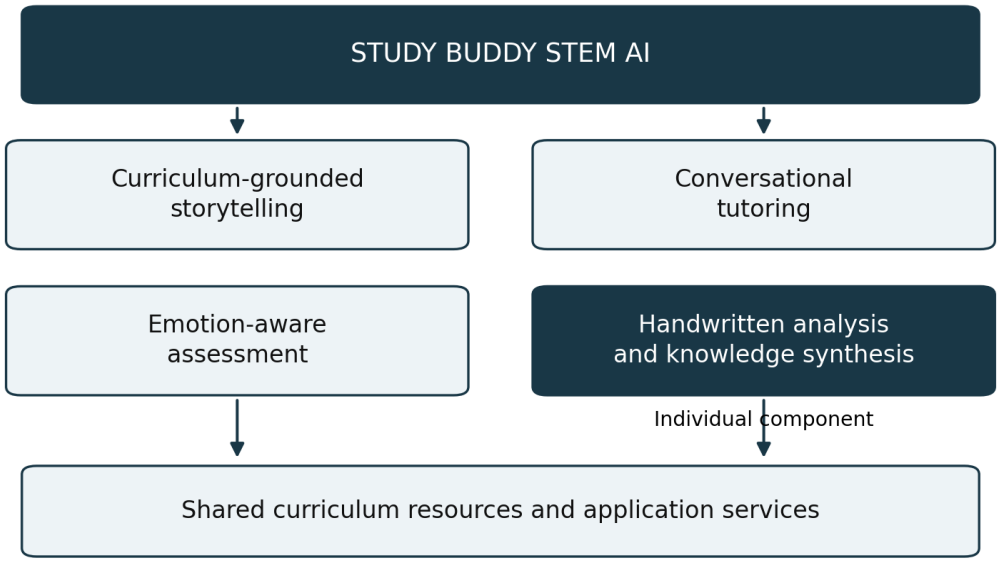


Figure 1.1: Individual component within Study Buddy STEM AI

1.2 Individual Component Context

The Handwritten Content Analysis and Knowledge Synthesis component accepts a photograph of a handwritten note or editable English text. Gemma 4 Cloud performs optical character recognition (OCR). The student can review the extracted wording before the system treats it as the material to be checked. A local Llama 3.2 3B model, accessed through Ollama, compares factual statements with retrieved curriculum evidence and produces a corrected note. Gemma 4 Cloud supports the other generation tasks, including flashcards, mind maps and structured revision notes. This division of responsibilities defines the configuration discussed throughout the thesis.

The component is a hybrid cloud and local system. Images submitted for cloud OCR leave the local processing boundary, and text used for cloud synthesis also crosses that boundary. Local verification reduces the need to send the verification workload to a remote model, but it does not make the complete workflow offline or local-only. These distinctions affect consent, operation during network failures, and the interpretation of deployment costs.

The technical component description records a FastAPI module, a shared Facebook AI Similarity Search (FAISS) retrieval index, claim-level evidence records, conservative note reconstruction, an optional repair-and-recheck path, and multimedia revision workflows [2]. These records establish the intended software

behaviour. They do not, by themselves, establish measured accuracy or educational benefit.

1.3 Problem Statement

The research problem is how to transform informal English science notes into useful revision resources while controlling the propagation of unsupported or incorrect statements. Three linked difficulties arise. First, handwriting recognition can change the meaning of a note before verification begins. Second, retrieved text may be related to a claim without supporting or contradicting it. Third, a generator may introduce an additional fact while rewriting a correct sentence or creating a study aid.

A single overall judgement cannot adequately explain these failures. If a note contains ten correct statements and one misconception, the learner needs to identify the misconception and understand the evidence for a correction. Conversely, a scientifically reasonable statement that is absent from the indexed material should not automatically be labelled false. A useful system must distinguish a content judgement from missing evidence and from an operational failure.

1.4 Research Gap

Retrieval-augmented generation (RAG) combines language generation with retrieved external information [3]. Evidence-based claim verification is also established: FEVER explicitly distinguishes supported, refuted and insufficiently evidenced claims [4]. Accordingly, this thesis does not claim to invent RAG, OCR, fact checking or automatic study-resource generation.

The gap investigated here is the integration and evaluation of these methods around student-authored English notes in a bounded Grade 10-11 science context. The proposed contribution connects claim-level feedback to note repair and then uses the repaired content as the input for revision resources. Its distinctive research question is whether maintaining that evidence chain offers practical advantages over directly rewriting the unverified note. This is a scoped research proposition, not a claim that no comparable system exists anywhere.

1.5 Aim and Objectives

The aim is to design and document an English-language note-analysis component that produces inspectable, curriculum-grounded feedback and evidence-constrained revision material. Table 1.1 connects each objective to an observable output and the evidence needed to assess it.

Table 1.1: Individual research objectives

Objective	Intended output	Evaluation evidence
O1 Capture English notes	Editable OCR transcript with uncertainty cues	Character and word errors against human transcription
O2 Check factual claims	Claim labels with supporting passages	Teacher labels and evidence annotations
O3 Reconstruct reliable notes	Accepted claims and rechecked corrections	Retention and unsupported-addition analysis
O4 Produce revision resources	Flashcards, mind maps and structured notes	Teacher review of correctness and coverage
O5 Assess practical operation	Traceable requests and separate failure states	Latency, failure and usability observations

1.6 Research Questions or Hypotheses

RQ1 asks whether curriculum retrieval improves claim-verification quality compared with the same local model without retrieved evidence. RQ2 asks whether selective, approved-source web fallback increases useful decision coverage without an unacceptable increase in unsupported decisions. RQ3 asks whether conservative reconstruction and independently rechecked repairs preserve valid information more reliably than direct whole-note rewriting. RQ4 asks whether resources generated from verified notes are more faithful and educationally usable than resources generated directly from raw notes.

1.7 Scope Limitations and Assumptions

The scope is English input and English output for Grade 10-11 science. Multilingual capabilities elsewhere in Study Buddy are outside this component's evaluation. Typed notes and lecture transcripts may enter the same text-processing pathway, but

handwriting remains the central use case. Recognition of arbitrary diagrams, complete mathematical proofs and unrestricted subjects is not an established capability.

The design assumes access to a versioned curriculum collection, permission to use the source material, a reachable cloud model for OCR and synthesis, and a local runtime for Llama verification. It also assumes that a teacher can interpret the relevant curriculum when producing reference labels. These assumptions must be checked before a classroom study. The system is a revision assistant and does not replace teacher marking or certify that a student's whole understanding is correct.

1.8 Individual Contribution

My individual contribution is the design and documentation of the note-analysis workflow: capture and review, claim extraction, curriculum retrieval, evidence-gated decisions, corrected-note preparation, and subsequent knowledge synthesis. The boundary includes the component interfaces and its evaluation plan. It excludes authorship of the foundation models, the shared conversational tutor, the storytelling module and the emotion-aware assessment module.

1.9 Thesis Structure

Chapter 2 reviews relevant methods and identifies the conceptual basis of the component. Chapter 3 defines the research design and evaluation procedure. Chapter 4 explains the solution architecture and design decisions. Chapter 5 documents implementation responsibilities, integration and testing. Chapter 6 establishes the current validation status and the reporting structure for future measurements. Chapter 7 discusses the implications and limitations. Chapter 8 considers commercialisation and sustainability. Chapter 9 concludes the thesis and sets out the next research steps. The appendices contain reusable evaluation instruments and the academic-integrity disclosure.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This chapter uses a targeted narrative review rather than a systematic review. Sources were selected for their direct relationship to retrieval, factual verification, educational feedback, knowledge representation, model deployment and evaluation. Primary research publications and official model or service documentation were preferred. The review does not report an exhaustive search, a PRISMA screening process or a count of all available products.

Table 2.1 distinguishes established methods from the specific note-analysis problem. Results from other datasets are not transferred numerically to this project.

Table 2.1: Related work and its relevance to the component

Source	Established contribution	Relationship to this research
Lewis et al. [3]	Retrieval-supported language generation	Basis for grounding; not a note-repair evaluation
Thorne et al. [4]	Claims, evidence and an insufficient-information label	Basis for explicit abstention
Glockner et al. [5]	Limitations caused by missing counter-evidence	Motivation to avoid treating source absence as falsehood
Daheim et al. [6]	Verification-informed feedback for reasoning errors	Supports separating error detection from feedback generation
Olney et al. [7]	Concept-map-based educational question generation	Related to coordinated maps and study questions
Es et al. [8]	Evaluation of retrieval-augmented generation	Supports evaluating retrieval and generation separately

2.2 Domain and Theoretical Foundations

2.2.1 Student notes as evidence of understanding

A student's note is both a learning resource and a record of an interpretation. It may compress a longer classroom explanation, omit a condition that seemed obvious during a lesson, or combine several ideas in one sentence. These properties make note analysis different from checking a short answer against a fixed examination rubric. The system must first identify what the student is asserting before deciding whether the assertion agrees with a source.

For this project, a factual claim is a statement that can be assessed against available science evidence. A heading, personal reminder or question is not automatically a factual claim. This distinction matters because counting every line as a right-or-wrong answer would distort the feedback. A fragment may also need its heading to resolve the subject. The representation must preserve that context without inventing information the learner did not write.

2.2.2 Retrieval and generation

Lewis et al. describe RAG as combining a generative model with retrieved information held outside its parameters [3]. The transferable idea is access to inspectable external material. Study Buddy uses this principle at the level of individual note claims. The project does not reproduce the original paper's training procedure or claim to achieve its benchmark results.

Retrieval and verification remain separate tasks. A retrieval score expresses ranking relevance under a particular method; it is not a probability that a claim is true. A passage containing the same scientific terms can discuss a contrasting case. A verifier must therefore consider negation, quantities, conditions and relationships, rather than accepting the highest-ranked text as confirmation.

2.2.3 Evidence and abstention

FEVER formalises a distinction between support, refutation and insufficient information [4]. This provides a stronger basis than a forced binary label for bounded-corpus verification. The component adds an educationally motivated incomplete label and a separate verification-failed state. The latter is an operational result and must not be interpreted as a property of the student's knowledge.

Glockner, Hou and Gurevych examine how missing or unrealistic counter-evidence limits fact-checking systems [5]. In this project, their work motivates caution when the retrieved curriculum does not settle a claim. It does not prove that an approved-domain fallback will solve the problem. That fallback is a design choice whose benefits and risks require separate measurement.

2.3 Related Research

2.3.1 Semantic retrieval and sentence embeddings

Sentence-BERT supports efficient semantic comparison by producing sentence representations that can be compared directly [9]. In a note-analysis setting, this is useful when a learner expresses an idea differently from the textbook. Semantic similarity alone, however, may not adequately distinguish a correct relationship from its reversal. This limitation motivates a complementary lexical search and a later evidence judgement.

The all-MiniLM-L6-v2 model card describes a sentence-transformer model that maps text into 384-dimensional embeddings [10]. The component documentation identifies it as the shared local embedding option. The precise deployed embedding configuration must be recorded when evaluation is executed, because changing the encoder changes retrieval behaviour. A model name should not be treated as a permanent guarantee that every existing index was built using that encoder.

FAISS provides infrastructure for vector similarity search; Johnson, Douze and Jégou describe efficient large-scale search methods associated with that work [11]. In this project, FAISS is an indexing tool rather than a scientific reasoning model. A returned neighbour supplies candidate evidence, not a verdict. Its usefulness depends on the quality of the curriculum chunks, metadata and embedding consistency.

2.3.2 Lexical retrieval and rank fusion

BM25 is a lexical relevance method within the probabilistic relevance framework discussed by Robertson and Zaragoza [12]. Exact terminology can be important in science notes: a small change in a unit, chemical symbol or technical noun may materially change a claim. The project therefore combines lexical retrieval with semantic retrieval instead of assuming that one ranking method captures every useful match.

Reciprocal rank fusion combines ranked lists through their positions rather than directly comparing incompatible raw scores. Cormack, Clarke and Büttcher present this method as an effective way to fuse retrieval results [13]. The component adopts the approach to merge dense and lexical candidates. Its parameter choice and

candidate limits remain engineering settings, not empirically optimal values for this corpus until tested.

A cross-encoder reranker can score a query and candidate passage together. The official BGE reranker documentation describes such relevance scoring [14]. The project keeps reranking optional because it introduces an additional model and resource requirement. Whether that cost improves retrieval enough to justify deployment should be determined by the ablation procedure, not by the presence of a more complex model alone.

2.3.3 Verification with external evidence

The AVeriTeC shared task evaluates verification with attention to both verdict quality and retrieved evidence quality [15]. This is relevant because a correct-looking label supported by an irrelevant citation is not a satisfactory result. The component's planned evaluation consequently includes citation adequacy as well as label agreement. Its approved-source web fallback has a narrower purpose than open-domain fact checking: it is invoked only after curriculum evidence cannot support a decision.

There is an important difference between syntactic citation validity and semantic support. Checking that S2 is among the retrieved evidence identifiers prevents the model from inventing that identifier. It does not prove that passage S2 entails the claim. Teacher annotation and error analysis are therefore necessary even when every response contains a technically valid reference.

2.3.4 Verification-informed educational feedback

Daheim et al. investigate verification of student reasoning errors and show how verification can inform more targeted tutoring responses [6]. Their setting involves reasoning chains rather than informal note photographs. The relevance here is the separation of identifying a problem from generating the response to it. The thesis applies that separation to a note-reconstruction workflow, where proposed corrections undergo a fresh check before inclusion.

2.3.5 Knowledge synthesis and retrieval practice

Olney, Graesser and Person describe question generation from concept maps extracted from textbook material [7]. Their work connects knowledge representation with educational questioning. In Study Buddy, the mind map and flashcards are alternative representations of a verified note. The project must still evaluate the accuracy of each relationship and answer; sharing a source does not automatically make the outputs mutually consistent.

Roediger and Karpicke provide evidence that retrieval practice can improve later retention under their experimental conditions [16]. This offers a rationale for flashcards that require a learner to recall an answer before revealing it. It does not establish that automatically generated cards in this project improve examination performance. Such a claim would require a separate learner study with an appropriate comparison and delayed measurement.

2.4 Existing Systems Products and Techniques

2.4.1 Cloud OCR and local text reasoning

The official Gemma 4 cloud listing describes a multimodal model capable of accepting images and text [17]. The project uses the name Gemma 4 Cloud for the OCR and synthesis service. No particular parameter count, quantisation, context window or commercial price is assumed for the deployed endpoint without a recorded configuration. The ability to accept an image is also not equivalent to verified accuracy on handwritten science notation.

Ollama's Llama 3.2 listing identifies its 3B variant as a text-input and text-output model [18]. This supports the separation of OCR from local text verification. It would be inaccurate to describe this 3B model as the image-recognition stage. The selected workflow passes reviewed text and evidence to the local model rather than sending it raw note images.

2.4.2 Study tools and multimedia resources

Standalone OCR, note rewriting, flashcard generation and video search serve useful but different purposes. Their integration can reduce repeated copying between tools,

but it also creates opportunities for errors to travel between stages. The design therefore treats the accepted note as a controlled handover point. The original note remains available for comparison, while unresolved claims remain outside the study-material source.

The YouTube search API supports filters such as SafeSearch and embeddability [19]. These filters help constrain discovery but do not certify the scientific correctness of a video. A search result, a short transcript summary and a curriculum-checked transcript are distinct outputs and must be labelled accordingly. Recommendation quality depends on what is searched and on the availability and quality of captions.

2.5 Gap Synthesis and Conceptual Framework

The literature supports the component methods more strongly than it supports the complete proposed workflow. The integrated framework in Figure 2.1 makes three relationships explicit: OCR quality affects claim interpretation; retrieval quality constrains verification; and the accepted claim set constrains subsequent generation. Failures can be studied at each boundary instead of being hidden within one overall satisfaction score.

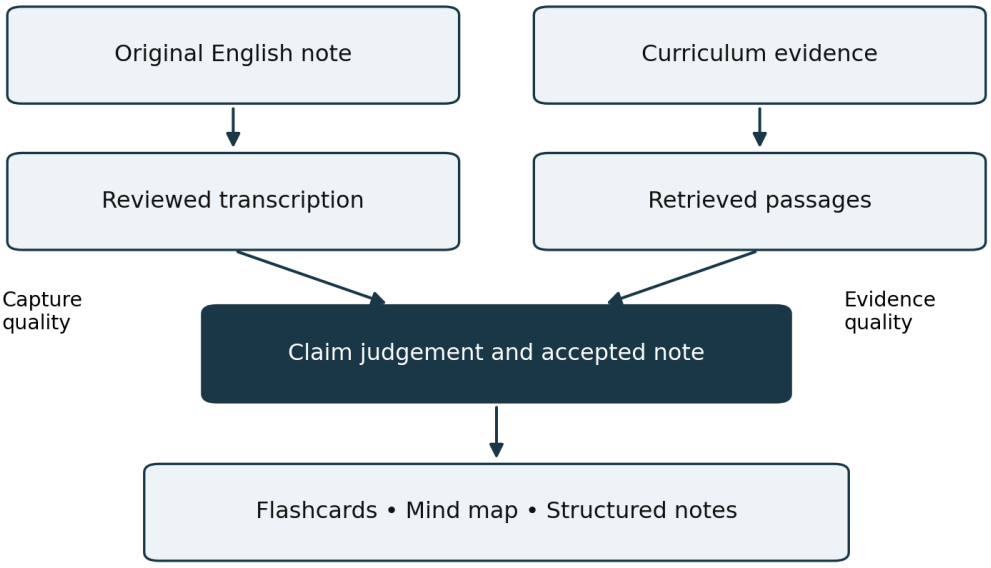


Figure 2.1: Evidence chain and evaluation boundaries

3 RESEARCH METHODOLOGY

3.1 Research Design

The study follows an artefact-oriented research design. The artefact is a note-analysis component with an explicit evidence policy, model allocation and set of user-facing outputs. Its design is informed by the problems identified in Chapter 1. The evaluation is intended to determine whether the artefact's behaviour supports the stated objectives, rather than merely whether its endpoints return valid responses.

Two forms of assessment are separated. Software verification concerns input handling, response schemas, failure states and integration contracts. Research validation concerns OCR fidelity, claim interpretation, evidence retrieval, verdict quality, note reconstruction and educational usability. Passing a schema test is necessary for dependable software, but it does not show that a claim is scientifically correct. Likewise, a convincing demonstration cannot replace an annotated evaluation set.

The study is planned as a staged evaluation. First, the corpus and configuration are frozen. Next, the stage-specific methods are evaluated using human-reviewed inputs. Finally, the end-to-end pipeline is assessed on original note images. This order allows an OCR error to be distinguished from a retrieval error and prevents an aggregate score from concealing the cause of a failed result. Figure 3.1 shows the proposed sequence; it is a study design, not a record of completed experiments.

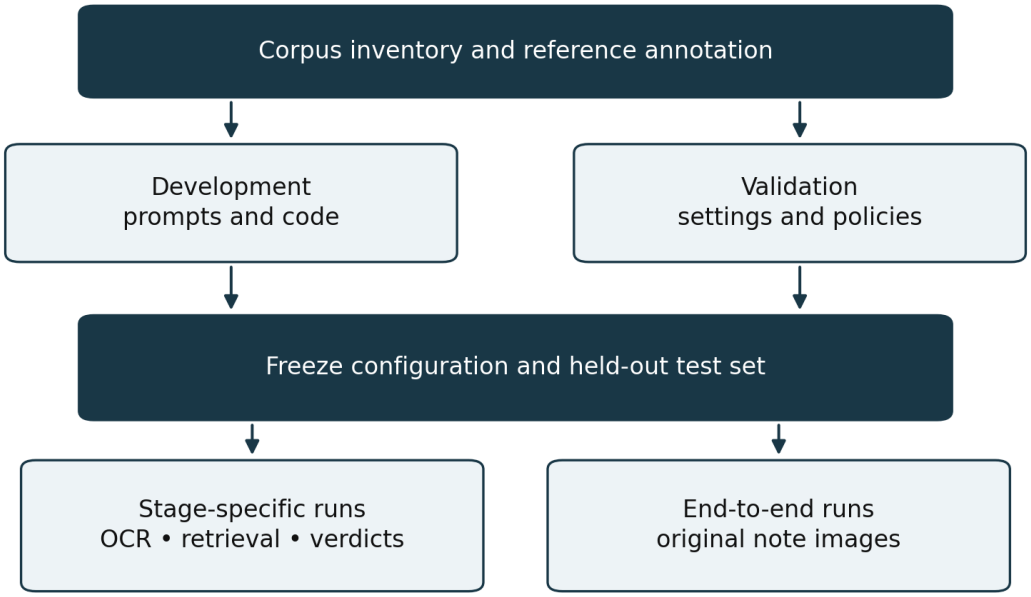


Figure 3.1: Planned evaluation sequence and separation of data

3.2 Data Participants or Study Materials

3.2.1 Curriculum corpus

The reference corpus is intended to contain the approved English Grade 10 and Grade 11 science materials used by the project. Each source needs a stable document identifier, title, grade, edition or year, page mapping and permission record. A curriculum label alone is not sufficient provenance. A later reviewer must be able to open the same page that informed the decision and determine whether the quoted passage was extracted faithfully.

Text extraction must preserve relationships that may be distributed across headings, captions and tables. Chunking should avoid separating a definition from its qualifying condition. When tables are linearised, row and column context must remain interpretable. Image-only pages require special handling; a missing extraction should be recorded as an ingestion limitation rather than silently treated as an empty topic.

3.2.2 English note collection

The proposed note set includes handwritten science notes with variation in handwriting, image quality, notation and topic. Typed transcriptions provide a separate reference for the OCR stage. The collection should include readable examples as well as difficult but interpretable cases; selecting only clean images would underestimate practical error. Completely unreadable content should be identified as such, not assigned an invented transcription.

Natural student notes and deliberately constructed diagnostic examples should be labelled separately. Diagnostic examples are useful for testing negation, omitted conditions, unit errors and ambiguous wording. They do not represent the prevalence of those errors in a classroom. If both types are included in the final benchmark, results should be reported by origin so that a balanced diagnostic set is not confused with a representative sample.

All collected material remains English-only. Scientific symbols and established formulae are retained as part of English science notation. Student names, school identifiers and unrelated personal writing should be removed or masked before research use. Recruitment, consent and any required institutional approval must be completed before collecting material from minors.

3.2.3 Annotation and reference labels

Teachers or appropriately qualified science reviewers should annotate the reference transcription, atomic claims, verdicts and supporting evidence. Annotation guidelines should define when missing information makes a claim incomplete and when the available sources are insufficient to decide. These categories are easy to confuse unless the intended curriculum context is made explicit.

At least two reviewers should independently annotate an agreed subset before discussing disagreements. Their initial labels must be preserved so that agreement can be measured. Adjudication then produces the final reference labels used for scoring. The number and qualifications of reviewers, the subset size and the adjudication process should accompany any reported metric. No completed teacher review is claimed at the present stage.

The reference record should also identify accepted alternative wording. A corrected sentence need not match one teacher's wording exactly to be valid. The assessment should focus on preserved meaning, missing conditions and unsupported additions. Appendix B provides the annotation form and label guidance.

3.3 Experimental or Study Procedure

3.3.1 Data separation and configuration freeze

Development, validation and test partitions should be separated before tuning the prompts or retrieval settings. Images and transcriptions of the same note must remain in one partition. Where feasible, related notes from the same learner should also remain together. Otherwise, the test set could contain close variants of material already used to refine the system.

The validation set is used to choose thresholds, prompt versions, candidate limits and optional calibration settings. The final test set is reserved for reporting. Repeatedly inspecting test failures and then adjusting the pipeline turns the test set into development data. If that occurs, a new untouched test set or an explicitly exploratory report is needed.

For every run, the record should include the local model identifier and digest, cloud endpoint identifier, generation settings, embedding configuration, index hash, reranker status, dependency versions and hardware. A cloud model may change behind a stable service name; the execution date and any provider revision should therefore be retained. Reproducibility means recording the environment actually used, not filling unknown settings with typical defaults.

3.3.2 Stage-specific and end-to-end runs

OCR is assessed first by comparing the recognised text with a human transcription. The verification stage is then assessed using the human transcription, which removes OCR error from that comparison. An additional run uses the actual OCR output, showing how recognition errors affect the full pipeline. The difference between these runs indicates the sensitivity of later stages to capture quality.

Retrieval is evaluated with gold claims and reference evidence. Verification is assessed twice: once with reference evidence and once with retrieved evidence. This

separates reasoning difficulty from retrieval difficulty. A model that performs well with teacher-selected passages but poorly with retrieved passages suggests a retrieval problem; poor performance in both settings suggests that the prompt, label definitions or reasoning model also require attention.

The end-to-end run begins with the original image and includes user review as a separately recorded condition. A fully automatic run and a reviewed-OCR run should not be combined without distinction. Human corrections can materially improve the input, so the analysis must state when they were permitted and how much time they required.

3.3.3 Baselines and ablations

The baselines in Table 3.1 are proposed comparisons. They use the same evaluation material and, where possible, the same model and generation settings. This reduces the risk of attributing a gain to retrieval when it actually arose from changing the model or giving one system more context.

Table 3.1: Planned baselines and ablation conditions

Condition	Controlled change	Main question
Ungrounded verification	No retrieved passages	Does evidence access improve verdicts?
Dense retrieval only	Remove lexical search and fusion	Does hybrid retrieval recover more evidence?
Curriculum only	Disable web fallback	What does fallback add or harm?
Hybrid without reranking	Keep fusion and remove reranker	Is reranking worth its latency?
Direct note rewriting	Rewrite the raw note in one step	Does claim-level control reduce unsupported content?
Conservative reconstruction	Keep only supported claims	How much valid content is retained?
Repair and recheck	Propose corrections and verify again	Can coverage improve without unreliable additions?
Raw-note synthesis	Generate aids before verification	Does the accepted note improve output faithfulness?

Ablations should remove one relevant mechanism at a time. Removing retrieval, changing the model and modifying the output format simultaneously would prevent a clear interpretation. Web comparisons also require comparable evidence snapshots or

time windows because search results may change. Any baseline that fails to run must be reported as a failed condition, not silently omitted from the sample count.

3.4 Measures and Evaluation Metrics

3.4.1 OCR and claim extraction

Character error rate (CER) and word error rate (WER) measure edit operations needed to match the reference transcription. Equation 3.1 gives the shared structure. S, D and I denote substitutions, deletions and insertions; N is the reference length measured in characters for CER or words for WER. Scores may exceed one when insertions are numerous. Empty references require an explicit handling rule.

$$\text{CER or WER} = (S + D + I) / N \quad (3.1)$$

Scientific notation should also receive a separate audit because a small character error can have a large effect on meaning. The audit records altered subscripts, superscripts, decimal points, units and signs. OCR confidence is not used as a substitute for measured transcription fidelity.

Claim extraction is assessed using teacher-defined claim boundaries and semantic matching rules. Precision measures how many extracted claims correspond to valid reference claims; recall measures how many reference claims were recovered. Duplicate and missed-claim rates are reported separately. A system should not improve its apparent verification score by extracting only the simplest statements.

3.4.2 Retrieval and verdict quality

Retrieval reporting includes Recall@k and mean reciprocal rank (MRR). Recall@k measures the proportion of annotated relevant evidence items retrieved within the first k results, averaged according to the declared protocol. If only the presence of any sufficient evidence is measured, that quantity should be called a hit rate rather than silently relabelled recall. Mean reciprocal rank uses the rank of the first relevant result and assigns zero when no relevant result is found within the evaluated list.

$$\text{MRR} = (1 / Q) \text{ sum over } q \text{ of } (1 / \text{rank}_q) \quad (3.2)$$

Verdict quality is measured with a confusion matrix and per-class precision, recall and F1 for the educational labels. Macro F1 gives each class equal weight.

Operational verification failures are reported separately and remain visible in the end-to-end outcome counts. A conditional F1 calculated only on completed requests must be accompanied by the completion rate so that frequent failures cannot disappear from the report.

3.4.3 Note reconstruction and synthesis

Reconstruction is assessed using valid-claim retention, correction validity, unresolved-claim handling and unsupported-addition rate. Retention compares the valid information preserved in the output with valid information identified in the original note. A short output can contain no false claims while still discarding most useful content. Reporting retention alongside correctness prevents that output from appearing unconditionally superior.

Flashcards are reviewed for answerability, correctness, ambiguity and coverage. Mind maps are reviewed at both node and relationship levels. Structured notes are reviewed for preserved conditions and absence of added facts. Table 3.2 identifies the principal measurement for each stage.

Table 3.2: Stage-specific evaluation measures

Stage	Main measure	Required companion information
OCR	CER and WER	Image quality and symbol errors
Extraction	Precision, recall and F1	Missed and duplicate claims
Retrieval	Recall@k and MRR	Corpus version and evidence relevance rules
Verification	Per-class and macro F1	Confusion matrix, completion and citation adequacy
Reconstruction	Valid-claim retention	Unsupported additions and unresolved content
Synthesis	Teacher rubric scores	Output type and source coverage
Operation	Median and tail latency	Hardware, network, request size and failures

3.4.4 Confidence and operational measures

A model-generated confidence number is not assumed to be calibrated. Guo et al. study calibration as agreement between predicted confidence and observed correctness [20]. Here, evidence-strength bands can be shown as provisional

indicators, while numerical probabilities require a separate validation procedure. The calibration plan must assess reliability on held-out data and state the class and event to which the probability refers.

Operational measurements include end-to-end latency, per-stage latency, failure rate and cloud usage. Median latency describes a typical request; an upper percentile such as the 95th percentile describes slower cases. Cold-start loading should be separated from warm inference. Response-speed ratings on an arbitrary scale are not interchangeable with seconds, and token counts are not a complete cost measurement without the applicable provider terms.

3.5 Data Analysis

The primary analysis compares matched inputs across conditions. Differences should be reported with uncertainty intervals and sample counts, rather than only a ranking of averages. Claims from one note are correlated. Resampling or paired comparisons should therefore use the note as the grouping unit where appropriate, instead of assuming every extracted claim is an independent student observation.

An error taxonomy supports interpretation of quantitative results. Categories include OCR corruption, missed claim, incorrect decomposition, absent evidence, irrelevant citation, wrong entailment judgement, unsupported repair and synthesis drift. A single input may have a chain of errors; the earliest identifiable cause and the final user-visible outcome should both be recorded.

Automated RAG evaluation frameworks, such as RAGAs, distinguish aspects of retrieval and generation quality [8]. They may assist exploratory analysis, but an automated judge should not be the sole reference for this educational study. If the judge is related to a model under evaluation, possible preference or shared-error effects must be acknowledged. Teacher review remains necessary for the central correctness questions.

3.6 Ethics Privacy Security and Responsible AI

The study involves educational material that may belong to minors. Before collection, the research team must establish the required institutional approval, school permission, guardian consent and learner assent processes. Participation

should be voluntary and separate from school grading. Declining participation should not prevent access to normal teaching. This section states the research safeguards to be implemented; it does not claim that approval has already been granted.

Cloud processing must be explained before upload. The consent information should distinguish image transfer for OCR from text transfer for synthesis. Local deletion of an uploaded file does not prove deletion from a provider's systems. Data handling must therefore reflect the actual provider arrangement, retention terms and application logs. No absolute claim of confidentiality is made on the basis of local inference alone.

UNESCO's guidance emphasises human agency and responsible educational use of generative AI [21]. In this component, the practical response is to preserve review opportunities, show uncertainty, avoid automated high-stakes grading and allow teachers to challenge an output. An English-only system also has a limited accessibility scope; multilingual support in another module does not remove that limitation.

3.7 Reproducibility and Research Artefacts

The study should retain the corpus inventory, anonymised annotation records, prompt versions, model configuration, evaluation scripts and run manifests. Sensitive note images require controlled storage and should not be published automatically with code. A public package can include de-identified diagnostic examples and aggregate results where permissions allow.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Functional requirements

The component must accept an English note, preserve a reviewable transcript, identify factual claims and return an evidence-aware judgement for each claim. A learner should be able to distinguish a supported statement from a corrected statement and from unresolved content. The corrected note must be available as the controlled source for subsequent revision resources. These requirements prioritise traceability over the appearance of a complete answer at every stage.

Image transcripts, typed notes and video captions retain their distinct provenance. Table 4.1 records the core functional requirements.

Table 4.1: Functional requirements and acceptance conditions

ID	Requirement	Intended acceptance condition
FR01	Capture English note images	Return editable text or a clear capture error
FR02	Review uncertain transcription	Preserve user corrections before verification
FR03	Extract atomic claims	Keep source links and avoid duplicate claims
FR04	Retrieve curriculum evidence	Return stable passage and page references
FR05	Gate educational verdicts	Require selected, valid evidence identifiers
FR06	Handle uncertainty and failure	Keep insufficient evidence separate from failure
FR07	Reconstruct and repair notes	Include supported or rechecked content only
FR08	Generate revision resources	Use accepted content and preserve its meaning
FR09	Support video learning	Distinguish discovery, summary and verification
FR10	Explain processing status	Show partial success and recoverable failures

4.1.2 Non-functional requirements

Reliability requires predictable failure behaviour. A timeout on one claim must not convert that claim into an incorrect answer or remove the other successful results. Privacy requires a clear account of which data reaches the cloud. Maintainability requires separate responsibilities for OCR, retrieval, verification and synthesis.

Usability requires evidence explanations that can be understood without knowledge of model internals.

4.2 Component Architecture Within the Complete Solution

The component registers its routes within the shared FastAPI application and reuses the project retrieval resources. The main boundary is the module and its data contracts, not a separate microservice. The frontend coordinates the user's upload, review, verification and synthesis actions. Figure 4.1 shows the local and cloud responsibilities. The diagram is an architectural representation, not a screenshot of an observed deployment.

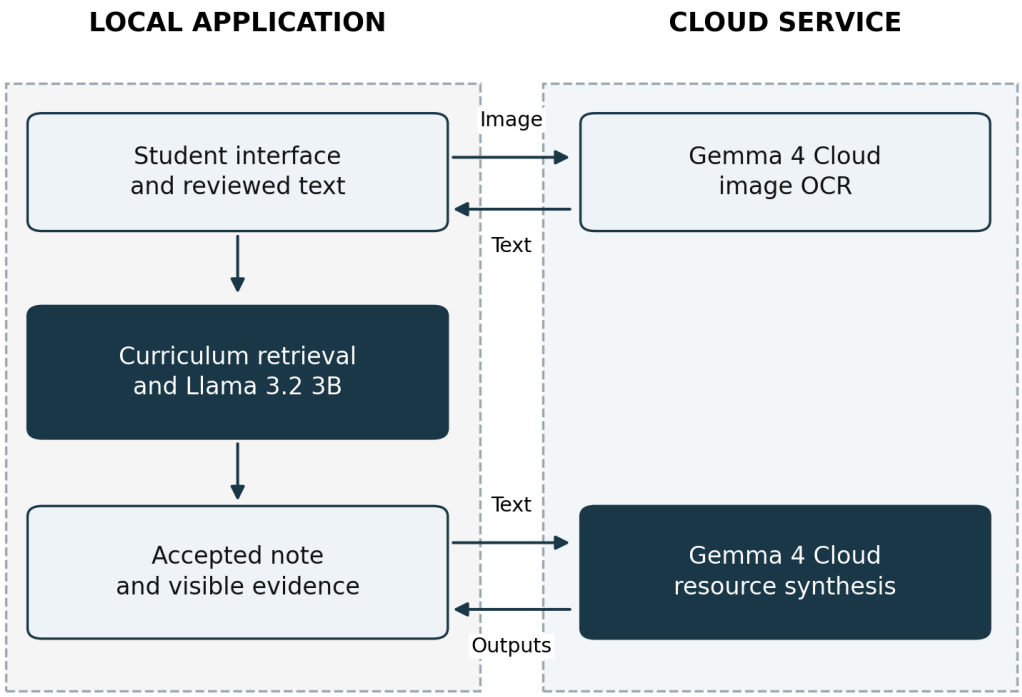


Figure 4.1: Hybrid cloud and local component architecture

The shared curriculum index is a dependency rather than a product output. If it is unavailable, the system must say that verification cannot proceed. It must not silently change to ungrounded answers while preserving the same verified label. This is an important integration contract because a user cannot otherwise distinguish a grounded result from a degraded fallback.

4.3 Design Alternatives and Decisions

The first alternative is direct whole-note rewriting. It has a simple interaction and may require fewer model calls, but it obscures the relationship between an original error and the resulting change. Claim-level checking is more expensive and can fragment context, yet it supports a more inspectable revision process. The proposed evaluation compares these alternatives rather than assuming that the additional complexity is always beneficial.

The second alternative is all-cloud processing. It would simplify local model deployment but increase dependence on remote inference for verification. An all-local design would reduce remote transfer but requires a suitable image-capable model and sufficient resources. The chosen hybrid allocation reflects the project's current configuration. It is not presented as the empirically cheapest or fastest architecture.

The explicit uncertainty and failure states avoid forced binary judgements. Table 4.2 summarises the principal alternatives and their trade-offs.

Table 4.2: Design alternatives and trade-offs

Decision	Reason for the selected design	Cost or remaining risk
Claims before rewriting	Locate changes and attach evidence	More calls and possible lost context
Hybrid retrieval	Combine semantic and exact-term matching	More configuration and evaluation
Curriculum before web	Preserve syllabus priority	Corpus gaps can reduce coverage
Cloud OCR with local checking	Separate image recognition from text reasoning	Network dependence remains
Explicit abstention	Avoid unsupported right-or-wrong decisions	Some notes remain incomplete
Repair with a fresh check	Test proposed wording before inclusion	Shared-model errors can persist

4.4 Detailed Design

4.4.1 Capture and OCR review

Image handling begins with validation of size and supported format. Orientation correction, contrast adjustment and downscaling are documented preprocessing

operations. Their effect must be evaluated because excessive transformation can remove fine notation. A decimal point or a faint minus sign may be more important than the visual neatness of the page.

Gemma 4 Cloud is instructed to transcribe the visible English text without silently correcting its science. The transcript should preserve line structure, units and formulae as far as possible. The review interface can identify regions such as paragraphs, equations and labels, and offer recovery suggestions for suspicious strings. Suggestions remain distinct from accepted edits. Changing a character based on a glossary is not the same as observing it in the image.

4.4.2 Claim extraction and source mapping

The design extracts up to 25 self-contained claims per request, as specified in the component notes [2]. Each claim retains its source fragment and location within the note. The limit bounds workload; it also means that longer notes require an explicit continuation or chunking policy. The interface must not imply that the entire note was checked if only the first set of claims was processed.

Atomicity is important. A sentence joined by “and” may contain two independently assessable statements. Splitting it allows one statement to be accepted while the other is corrected. However, a condition must remain attached to the claim it qualifies. A decomposition that drops “at constant temperature” or “in a closed system” can create an error that was not present in the original writing.

Exact-fragment mapping supports repair. Overlapping or duplicate mappings are rejected rather than edited automatically. Headings and list structure are preserved so that the corrected output remains recognisable to the learner. Questions and non-factual reminders are not included in the scored claim set.

4.4.3 Hybrid curriculum retrieval

For each claim, dense search and BM25 return candidate passages. The documented configuration retrieves up to 20 candidates from each branch, merges them using reciprocal rank fusion (RRF) with $k = 60$, and optionally reranks the fused candidates. The final evidence context contains up to five passages identified as S1 to S5. Figure 4.2 illustrates these operations.

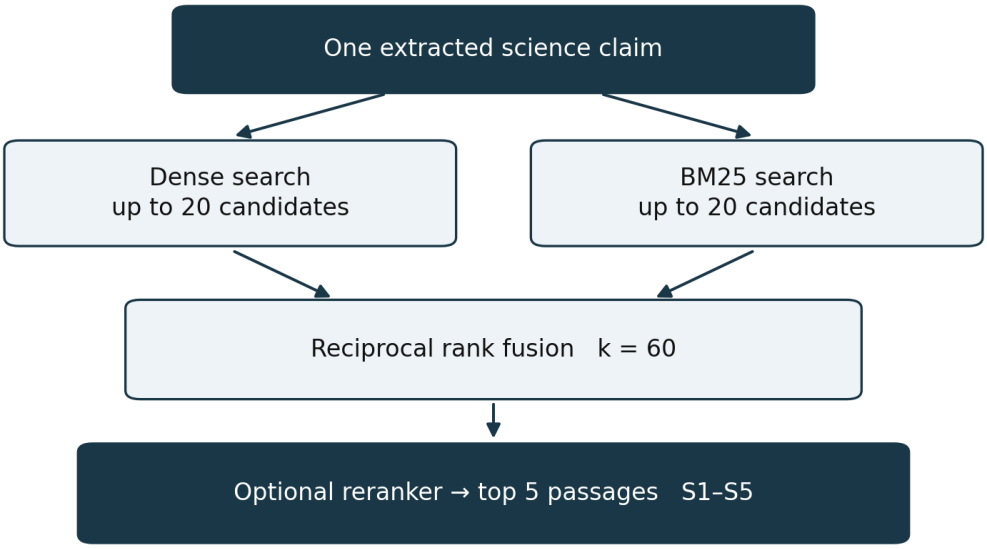


Figure 4.2: Claim-level hybrid retrieval and evidence selection

$$RRF(d) = \sum_{r} \frac{1}{60 + rank_r(d)} \quad (4.1)$$

In Equation 4.1, a document absent from a ranked list contributes zero for that list. Fusion ranks candidates; it does not establish truth. The optional reranker is allowed to change their order, while a clearly reported fallback to fusion order is used if reranking is unavailable. The health response should expose that difference because comparisons made under different retrieval modes are not equivalent.

Evidence records retain document, page, passage and retrieval metadata. Any similarity-threshold policy requires validation against the deployed corpus and retriever.

4.4.4 Evidence-gated verification

Local Llama 3.2 3B receives the claim, the evidence identifiers and an instruction to judge only what the evidence supports. Its structured response contains the verdict, explanation, selected evidence identifiers and an optional correction. Code validates the identifier selection before accepting an educational verdict. If the model names a nonexistent identifier or provides none, the claim cannot be marked supported solely on the basis of its fluent explanation.

The five states in Table 4.3 separate learner-content feedback from uncertainty and failure. Figure 4.3 shows their relationship to the evidence policy. A valid citation is

a necessary control, not a proof that the inference is correct; semantic adequacy remains part of the evaluation.

Table 4.3: Verification states and downstream treatment

State	Interpretation	Downstream treatment
Correct	Selected evidence supports the claim	Eligible for the accepted note
Incorrect	Selected evidence contradicts the claim	Repair and recheck, or exclude
Incomplete	Essential information is missing	Repair and recheck, or exclude
Insufficient evidence	Available material cannot settle the claim	Preserve as unresolved; do not penalise
Verification failed	A required operation did not complete	Show failure; permit a safe retry

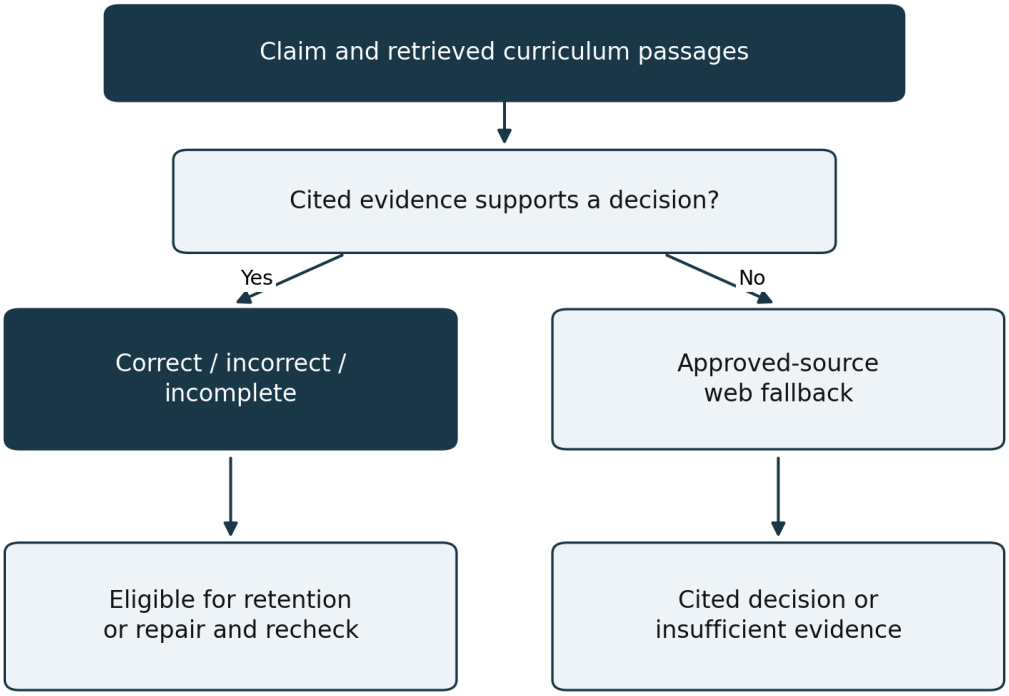


Figure 4.3: Evidence policy for decisions and abstention

4.4.5 Selective web fallback

Web fallback is invoked when the curriculum stage cannot make a cited decision. The component description specifies filtering search results through an approved-domain list. An empty list means that no web decision is permitted. A cited approved result may support a correct or incorrect judgement; otherwise the claim remains

insufficiently evidenced. Incomplete curriculum claims are not automatically promoted through this fallback path.

A domain allow-list restricts provenance but does not guarantee the adequacy of every snippet. The selected evidence may omit a qualification, contain outdated material or refer to a different phenomenon. The result should identify web evidence as distinct from curriculum evidence. The evaluator must inspect false upgrades as well as newly decidable claims. Increasing coverage by making unreliable decisions would defeat the purpose of the fallback.

4.4.6 Scoring and coverage

The note score uses only decidable claims. Let C be correct claims, I incorrect claims, P incomplete claims and T all processed claims. Equations 4.2 and 4.3 define the displayed score and coverage. If C + I + P is zero, the score is unavailable rather than zero. If T is zero, coverage is also unavailable.

$$\text{Note score} = C / (C + I + P) \quad (4.2)$$

$$\text{Decision coverage} = (C + I + P) / T \quad (4.3)$$

An illustrative calculation clarifies the distinction. If six claims are correct, two incorrect, one incomplete and three unresolved, the note score is 6/9 and coverage is 9/12. These are explanatory values, not project results. A high score with low coverage should never be displayed as a reliable judgement of the complete note. Incomplete claims currently receive no partial credit; that policy must be stated to the learner.

4.4.7 Conservative reconstruction and repair

Conservative reconstruction keeps claims whose final verdict is correct and whose evidence selection is valid. Incorrect and incomplete claims are excluded, while insufficient and failed claims are presented separately. Local Llama then rewrites the accepted content without adding facts. This approach can reduce unsupported material but may remove a useful idea instead of repairing it.

The repair pathway attempts to recover that information. It maps a claim to the source fragment, proposes a replacement using the cited evidence and runs fresh retrieval and verification on the replacement. Only a cited correct result is inserted.

Figure 4.4 shows the acceptance gate. A second model call is operationally separate, but it is not statistically independent when the same model and evidence are used. Correlated reasoning errors remain possible.

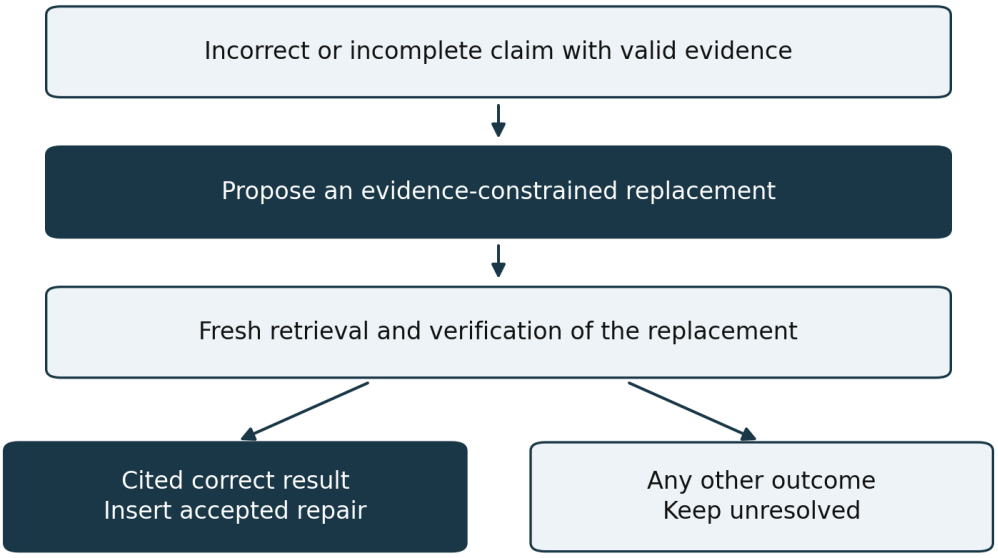


Figure 4.4: Proposed correction and re-verification gate

Original text and revision history remain available. Undo should not silently restore a known error into the accepted study source. The design instead separates the original view from the accepted-content view. This distinction allows a learner to inspect an earlier sentence without accidentally using it as verified material for flashcards.

4.4.8 Knowledge synthesis

Gemma 4 Cloud generates flashcards, a mind map and structured notes from the accepted text. The source handover includes the approved content and its identifiers where supported. The prompt limits generation to that content. Because instructions are not a mathematical guarantee, the resulting resources require their own faithfulness checks.

Flashcards should ask one clear question with an answer supported by the source. A mind map represents concepts as nodes and meaningful relationships as labelled edges. Structural sanitation removes edges pointing to nonexistent nodes, but it cannot establish that an edge is scientifically correct. Structured notes organise the

same content into headings, concise explanations and key terms. Figure 4.5 shows how the outputs share a source while remaining separately reviewable.

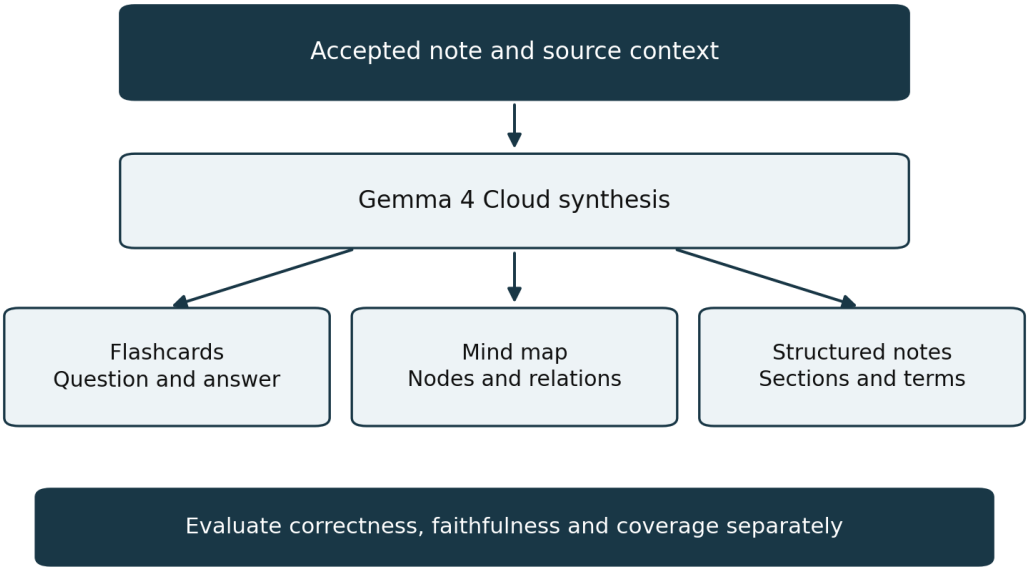


Figure 4.5: Accepted content and separate synthesis outputs

4.4.9 Video discovery and transcript analysis

Topic tags extracted from the corrected note guide video discovery. This reduces the risk of using a student's misconception as a search phrase. Discovery returns candidate learning resources and should not claim they have been verified. The short video-summary pathway summarises available captions without a full curriculum check. The full analysis pathway sends transcript content through verification and synthesis.

Caption availability, language and length limit the analysis. A preview or truncated caption segment is not evidence about an entire lecture. Long-video chunking and complete coverage tracking remain future work in the component notes. The interface should identify the analysed segment and disclose missing captions rather than presenting a partial summary as a comprehensive validation.

4.5 Standards and Professional Practice

Traceable identifiers, versioned configuration and accessible controls are required before classroom deployment.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The component description places the backend under app/m4 and its interface panels under web within the shared application [2]. FastAPI handles application programming interface (API) routes and Pydantic models define request and response structure. Retrieval reuses the curriculum index and embedding configuration. The frontend uses JavaScript and D3 for mind-map interaction. These are documented implementation choices; runtime versions and hardware must be captured with the executed evaluation.

The confirmed model allocation is recorded in Table 5.1. Stage-specific routing takes precedence over earlier single-provider descriptions.

Table 5.1: Technology and model responsibilities

Layer	Technology or model	Responsibility
API	FastAPI and Pydantic	Validation, routing and structured responses
OCR	Gemma 4 Cloud	English handwriting transcription and capture support
Verification	Llama 3.2 3B through local Ollama	Compare claims with evidence and propose note corrections
Dense retrieval	FAISS and configured embeddings	Retrieve semantically related curriculum passages
Lexical retrieval	BM25 and RRF	Retrieve exact terms and combine ranked candidates
Optional reranking	BGE reranker	Reorder candidate evidence when available
Synthesis	Gemma 4 Cloud	Flashcards, mind maps, notes and other generation tasks
Interface	JavaScript and D3	Review, evidence display and graph interaction

5.2 Component Implementation

5.2.1 Input and OCR processing

The documented upload interface accepts common image formats with a 10 MB limit. Client-provided file type information should not be the only validation

boundary. The component notes contain differing statements about file-signature validation, so robust server-side content validation remains an explicit verification item rather than an assumed completed safeguard. The final deployment needs to reject unsupported content before it reaches an image decoder or provider.

An accepted transcript receives a revision identifier. Subsequent verification should use that exact revision. If a user edits the text after verification, earlier claims and verdicts become stale. The interface should require re-verification or clearly mark previous results as belonging to the older text. This prevents evidence from appearing attached to a statement it never assessed.

5.2.2 Verification orchestration

The orchestration separates claim extraction, retrieval, model judgement, evidence validation and optional web fallback. Each claim is processed within its own failure boundary. A model timeout creates a verification-failed record for that claim and leaves successful records intact. Counts and coverage are computed from the final state list rather than from an earlier intermediate response.

The evidence layer resolves identifiers and page links through approved document records. It must not accept arbitrary filesystem paths from generated output. Excerpts shown to the learner should come from the stored retrieved passage, not from an unconstrained model paraphrase presented as a quotation. The explanatory paragraph may summarise the source, but its status as explanation should remain clear.

Confidence handling is separate from verdict selection. The component documentation describes optional calibration artefacts and retrieval-policy signatures. A stale calibration model should not be loaded merely because its file exists. The deployed retriever, reranker and revision must match the configuration for which the artefact was validated. Until the required evidence exists, calibrated probability claims remain out of scope.

5.2.3 Corrected-note and synthesis interfaces

The conservative note endpoint and the repair endpoint implement different policies. Their outputs should state which policy was used. A learner may prefer a shorter accepted note for quick revision or a structure-preserving repair for comparison with

the original. In either case, unresolved claims remain separately accessible and must not disappear without explanation.

The documented standalone synthesis endpoint does not itself require FAISS. This is an important boundary: evidence-constrained generation depends on supplying the accepted note through the intended workflow. A stronger future contract would accept verified claim identifiers or a server-issued note revision token and reconstruct the synthesis source on the server. That would make the evidence restriction less dependent on frontend behaviour.

5.2.4 API inventory

Table 5.2 presents the canonical route responsibilities documented for the component. Legacy aliases may support older interface calls, but the namespaced paths should be used in new integration work. An API response should include enough state for the frontend to distinguish an empty result, an unresolved result and a failed request.

Table 5.2: Canonical component API responsibilities

Method and route	Purpose
GET /api/m4/health	Report provider, index and retrieval readiness
GET /api/m4/documents/{id}	Resolve an approved curriculum document
POST /api/m4/upload-ocr	Convert an image to editable text
POST /api/m4/ocr-review	Return text with region and uncertainty information
POST /api/m4/verify	Return claim verdicts, evidence and coverage
POST /api/m4/correct-note	Reconstruct from supported claims
POST /api/m4/repair-note	Propose and independently recheck corrections
POST /api/m4/synthesize	Generate flashcards, a mind map or structured notes
POST /api/m4/youtube-suggest	Discover candidate learning videos
POST /api/m4/video-summary	Produce a lightweight caption summary
POST /api/m4/youtube	Analyse captions through verification and synthesis

5.3 Integration With the Other Components

The individual component relies on shared curriculum access and connects to the wider learning interface. A mind-map concept can initiate a question in the core tutoring panel, while a student returning from an assessment can inspect a related note. These are workflow connections. They do not imply that this component trains the assessment model or creates the storytelling persona classifier.

Integration must preserve the distinction between the source of a learning difficulty and an observed system output. An incorrect claim in a note may indicate a misconception, an OCR mistake or an incomplete sentence. It should not automatically become a permanent learner weakness in another component. A future learner-profile integration needs an explicit provenance and confirmation policy before such observations influence adaptive assessment.

Figure 5.1 describes the request lifecycle and its recovery points. It shows why partial success is valuable: a synthesis failure does not invalidate an already produced verification result. Conversely, a successfully generated flashcard does not retroactively validate the note from which it was generated.

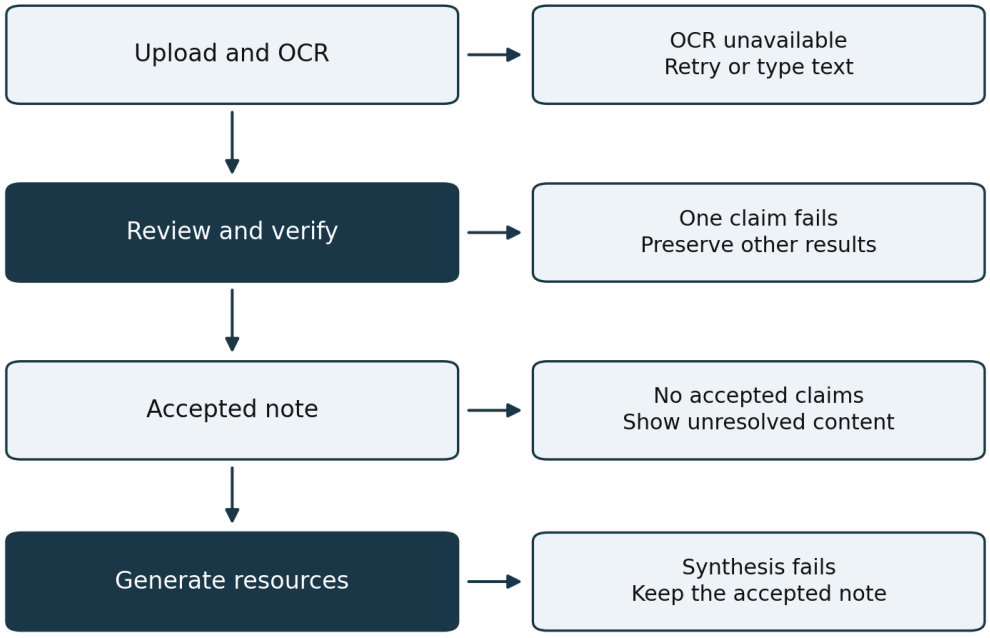


Figure 5.1: Request lifecycle and recoverable failures

Shared-resource changes require integration checks: a new encoder may invalidate an index, and revised document metadata may break page links. Record the shared configuration with each run.

5.4 Verification and Software Testing

5.4.1 Test strategy

The test strategy progresses from unit checks to contract checks, integration checks and end-to-end scenarios. Unit tests can verify score arithmetic and evidence-ID validation without calling a model. Contract tests can check response shapes and error mapping using controlled provider responses. Integration tests exercise the real index and provider connections. End-to-end tests begin with an image and inspect the resulting accepted note and resources.

The component notes mention tests and scripts, but no executed logs or test artefacts accompany the current thesis. Consequently, this chapter specifies expected behaviour and a test catalogue rather than claiming a pass count. The tests in Appendix C have status “Not executed in this report.” When executed, each requires an actual output, date, configuration and pass/fail decision supported by evidence.

5.4.2 Critical negative tests

Negative tests are particularly important for an evidence-aware workflow. A fabricated citation identifier must not be accepted. An empty approved-domain list must not allow a web-based upgrade. A model timeout must not lower the student's note score as though the student wrote an incorrect fact. A corrected sentence without sufficient support must remain excluded even when it reads fluently.

Input tests should include oversized files, corrupted images, unsupported content, empty text and a note exceeding the claim limit. Retrieval tests should include a missing index, irrelevant top results and a reranker failure. Interface tests should include editing text after verification and requesting synthesis while no claims have been accepted. These scenarios address actual failure boundaries in the design rather than merely mirroring function names.

5.4.3 Traceability and acceptance

Table 5.3 maps the most important requirements to tests and research measures. This makes it possible to distinguish an implementation defect from a scientific performance limitation. A valid-ID check can pass even when semantic citation adequacy is poor; both observations should remain visible.

Table 5.3: Requirement to verification traceability

Requirement	Software check	Research check
OCR review	Editable transcript and revision tracking	CER, WER and user correction burden
Evidence gate	Reject missing or invented IDs	Teacher-rated evidence adequacy
Abstention	Preserve insufficient-evidence status	False-rejection and coverage analysis
Score calculation	Exclude failed and unresolved states	Interpretation of note-level scores
Repair control	Require a fresh cited decision	Teacher acceptance of replacements
Synthesis restriction	Supply accepted content	Unsupported additions and source coverage
Partial failure	Preserve successful stage outputs	Failure rate and recovery usability

5.5 Deployment and Operation

Deployment requires local Ollama, a loaded curriculum index, cloud-service configuration and optional video-search credentials. Credentials belong in protected server configuration. Logs should retain operational identifiers and timings without unnecessary note text or images. Health information must not expose secrets.

Cloud outages prevent OCR and cloud synthesis, although typed-note verification may remain available locally. The interface should identify which functions still work. Full offline operation would require separately validated local OCR and synthesis. The documented flashcard schedule is in-page rather than persistent. Authentication, roles, rate limits and consent enforcement require implementation checks before broader deployment; a working demonstration alone does not establish production readiness.

6 RESULTS AND VALIDATION

6.1 Research Results

Component-level empirical evaluation has not yet been completed. There are therefore no measured OCR error rates, claim-verification scores, retrieval comparisons, teacher agreement statistics or learning outcomes to report. The research questions remain open. The present chapter records the validation status and the evidence required to turn the documented design into an evaluated research contribution.

6.2 Product and Technical Results

The current technical account establishes a defined component boundary, a hybrid model allocation, a retrieval-and-verification policy, a note-repair policy and a set of interfaces. These are design and documentation outcomes. They should not be interpreted as independently reproduced evidence that every route or safeguard operates as described. Table 6.1 distinguishes the present basis from the outstanding validation.

Table 6.1: Current evidence and validation status

Area	Present basis	Outstanding evidence
Model allocation	Confirmed cloud and local roles	Executed configuration and request traces
OCR	Documented capture and review pathway	Reference transcriptions and measured errors
Verification	Defined labels and evidence gate	Teacher-labelled benchmark and verdict outputs
Retrieval	Documented hybrid pipeline	Corpus inventory and retrieval results
Note repair	Defined proposal and recheck policy	Before-and-after outputs with teacher review
Synthesis	Defined study-resource formats	Faithfulness and educational-quality ratings
Integration	Documented shared API architecture	Reproducible end-to-end execution logs

6.3 User or Stakeholder Validation

No completed teacher or student validation is reported. The planned teacher review will assess factual decisions and the adequacy of corrections. A separate usability review will assess whether learners can understand the verdicts, inspect evidence and recognise unresolved content. These are different questions and require different instruments.

Preference and correctness are evaluated separately. Fluent feedback may be attractive without being supported, while accurate feedback may still be difficult for a learner to understand.

The proposed tasks include correcting an OCR error, locating a cited passage, explaining why a claim remains unresolved and checking a generated flashcard against the accepted note. Observed task completion, confusion and recovery attempts would be more informative than a single general question about whether the application is useful. Appendix E provides a concise usability protocol.

6.4 Integration Results

The integrated learning sequence is specified but has not been supported by executed integration artefacts in this report. Its acceptance conditions include preserving the accepted note when synthesis fails, keeping source provenance when a mind-map node opens the tutor, and preventing stale verdicts from being attached to edited text. Each condition needs an observed result from the deployed configuration.

6.5 Summary of Findings

The defensible finding at this stage is that the research has a specified evidence-aware workflow and a concrete evaluation design. The architecture addresses identifiable failure points through reviewed transcription, claim-level evidence, abstention and controlled reconstruction. Whether these mechanisms improve outcomes remains to be measured.

7 DISCUSSION

7.1 Interpretation of Findings

The main design insight is that reliability depends on the relationships between stages, not only on the capability of an individual language model. An accurate OCR transcript can still lead to a wrong verdict if retrieval misses the relevant condition. A correct verdict can still lead to an unreliable flashcard if synthesis adds information. The architecture makes these transitions visible and gives each one a corresponding evaluation question.

7.2 Comparison With Literature Baselines and Existing Solutions

The component adapts established methods to a specific learning workflow. Compared with general RAG, it introduces a student-note claim as the retrieval query and makes the resulting decision part of a reconstruction policy. Compared with claim-verification benchmarks, it continues beyond the verdict to a learner-facing corrected note and revision resources. Compared with direct resource generation, it restricts the intended source to accepted content.

These distinctions describe functionality and research scope. They do not establish superior accuracy or novelty in every aspect. Table 7.1 identifies the comparisons that would support stronger conclusions and the interpretations that must be avoided.

Table 7.1: Claims that require comparative evidence

Potential claim	Necessary comparison	Interpretation to avoid
Retrieval improves verification	Same model with and without evidence	Attributing a model change to RAG
Hybrid search improves evidence access	Dense-only versus hybrid on the same corpus	Treating similarity as truth probability
Repair improves note reliability	Direct rewrite versus repair and recheck	Ignoring valid content that was removed
Verified input improves study aids	Raw versus accepted input with matched generation	Assuming correct input prevents all hallucination
Hybrid deployment improves practicality	Measured latency, cost and availability	Assuming local means fully offline or private

7.3 Research Contribution

The proposed research contribution is an integrated, English-language framework for analysing student notes against a bounded curriculum and carrying evidence-aware decisions into note reconstruction and synthesis. The evidence policy is central: unsupported decisions remain unresolved, and proposed repairs must pass another check before being accepted.

7.4 Product Contribution

The product contribution is a coherent revision interaction. A learner can move from a handwritten note to a reviewed transcript, inspect claim-level feedback and then choose a study format. Preserving the original text alongside the accepted version helps the learner understand what changed. The value proposition is therefore more specific than automatic summarisation: it is an inspectable process for preparing revision material from personal notes.

7.5 Limitations and Threats to Validity

7.5.1 Construct and measurement validity

The meaning of an incomplete claim depends on context. A short note may intentionally omit detail that a full examination answer requires. Annotators need a shared standard so that the system is not penalised for preserving a legitimate shorthand or rewarded for adding unnecessary information. Rubric quality is therefore part of the evaluation, not an administrative detail.

7.5.2 Internal and external validity

Prompt tuning on test examples, near-duplicate notes across partitions and changing cloud configurations can weaken internal validity. Different model settings across baselines can create unfair comparisons. A controlled run manifest and a frozen test set reduce these risks but do not eliminate all nondeterminism.

External validity is limited by the selected grades, subjects, language, handwriting and capture conditions. Results on English science notes cannot be generalised to Sinhala notes, university mathematics or unrestricted diagrams without further study. Small teacher panels and convenience samples also limit the breadth of conclusions.

Reporting the actual sample characteristics is more useful than presenting the system as universally applicable.

7.5.3 Operational and educational limitations

Cloud access creates dependence on connectivity and provider availability. Local inference still requires adequate memory and processing capacity. A small local model may misinterpret a subtle condition even when the correct evidence is present. Repeated verification with the same model can preserve a correlated error rather than detect it.

7.6 Implications

For teachers, the framework could provide a structured starting point for discussing misconceptions, provided that its evidence and uncertainty remain visible. For learners, it could reduce the effort of turning personal notes into revision formats while preserving a link to the curriculum. For developers, it shows why provenance should be treated as a data relationship that persists across stages.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The proposed value proposition is to help English-medium Grade 10 and Grade 11 science learners turn their own notes into inspectable revision material. The student supplies the material already used in study; the system adds transcription, evidence-aware feedback and alternative revision formats. Teachers and tuition providers are potential supporting users because they can review the feedback and identify where automated interpretation is unreliable.

This is a proposed market position, not evidence of demand. No completed customer survey, willingness-to-pay study, school procurement agreement or revenue result is reported. Early validation should establish whether note checking solves a frequent enough problem and whether learners understand the difference between supported, incomplete and unresolved content.

8.2 Market and Competitive Position

The relevant alternatives include manual teacher feedback, standalone OCR, general-purpose chat tools and automated study-aid generators. The component's proposed differentiator is the visible connection between a student's statement, curriculum evidence and the resulting correction. A comparison should measure that benefit directly rather than relying on a claim that other products have no similar functionality.

A pilot should measure OCR-review effort, source-inspection behaviour and reuse of study materials. Attractive outputs alone do not establish that the workflow saves time.

8.3 Business or Sustainability Model

A possible delivery model is a school or tuition-centre pilot with a bounded curriculum collection and controlled access. A second possibility is an individual learner subscription with clear limits on cloud processing. These are alternatives for investigation, not established business arrangements. Advertising around learner data

would introduce additional privacy and governance concerns and is not assumed in the design.

Operating cost has several components: cloud OCR, cloud synthesis, local compute, storage, support and curriculum maintenance. Equation 8.1 gives a planning identity rather than a price estimate. Actual rates and usage must be recorded before calculating a budget.

$$\text{Monthly cost} = \text{cloud usage} + \text{hosting} + \text{storage} + \text{support} + \text{content maintenance} \tag{8.1}$$

8.4 Scalability Intellectual Property and Compliance

Scaling the system requires controlled concurrency, request limits, caching policies and monitoring. Shared curriculum embeddings can be reused, but private notes must not be exposed through an incorrectly scoped cache. Cloud quotas and local model queues can create different bottlenecks. A classroom pilot should establish peak-load behaviour before larger deployment is proposed.

The project uses existing models, libraries and educational sources. Their licences and provider terms must be reviewed for the actual deployment. Possession of a textbook PDF does not automatically grant permission to redistribute it through public citation links. Access control and source permissions are therefore part of the deployment design. This thesis does not assert legal clearance or certification.

Privacy arrangements should match the actual cloud and local data flow. A school-facing deployment needs clear responsibilities for consent, retention, deletion requests and incident handling. These arrangements require institutional review before release. The proposed stages in Figure 8.1 keep evidence collection and limited piloting ahead of wider deployment.

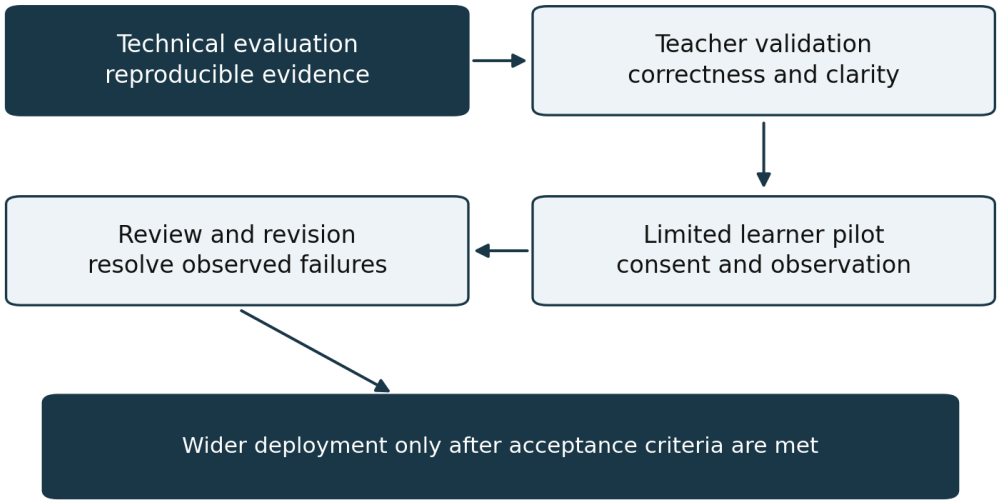


Figure 8.1: Proposed evidence-led deployment stages

8.5 Contribution to the Sustainable Development Goals

The project's intended educational purpose is aligned with the broad aim of improving access to useful learning support. Any contribution to Sustainable Development Goal 4 should be stated as an intended alignment rather than a measured social impact. The English-only scope and reliance on cloud access may exclude some learners, so access claims need to remain qualified.

Practical sustainability includes maintaining the curriculum corpus, minimising unnecessary model calls and supporting modest local hardware where feasible. Resource use should be measured before making environmental-efficiency claims. A smaller local model is not automatically the lowest-energy option if it requires repeated unsuccessful attempts. The product should prioritise useful, reliable outcomes and accountable maintenance over the volume of generated material.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presents the Handwritten Content Analysis and Knowledge Synthesis component of Study Buddy STEM AI for English Grade 10-11 science notes. The workflow connects cloud OCR, local curriculum-based claim verification, corrected-note generation and cloud-supported revision resources. Its central design principle is that unsupported content should not silently become accepted study material.

The architecture separates incorrect claims from incomplete claims, insufficient evidence and processing failures. It also distinguishes a valid citation identifier from genuinely adequate evidence. These distinctions make the proposed system more inspectable, but their practical benefit has not yet been established through component-level experiments. The thesis therefore documents a research artefact and its evaluation method without claiming unmeasured accuracy or learning improvement.

9.2 Achievement of Objectives

The design addresses the five objectives through defined capture, verification, reconstruction, synthesis and operational interfaces. Their empirical achievement remains to be assessed. Table 9.1 identifies the next evidence required for each objective, preserving a clear boundary between an implemented or documented mechanism and a validated outcome.

Table 9.1: Objectives and remaining validation

Objective	Present outcome	Next evidence
O1	English OCR and review design	Annotated image evaluation
O2	Claim and evidence policy	Teacher-labelled verification benchmark
O3	Reconstruction and repair rules	Retention and correction-quality results
O4	Three revision-resource formats	Faithfulness and usability review
O5	Instrumented evaluation plan	Executed latency and failure records

9.3 Individual Research and Product Contributions

The individual work defines the component workflow, its evidence controls, integration responsibilities and evaluation instruments. The use of established models and retrieval methods is acknowledged. The proposed contribution lies in their curriculum-bounded arrangement around learner-authored notes and in the testable relationships between verification, repair and synthesis.

9.4 Future Directions

The immediate priority is to freeze the deployed configuration and execute the stage-specific and end-to-end evaluations. Teacher annotation should precede threshold tuning and confidence calibration. Representative failures should guide improvements in source chunking, claim extraction, OCR review and repair prompts. Subsequent work includes enforcing accepted-source identifiers at synthesis, adding persistent notebooks and flashcard schedules, improving long-transcript coverage and evaluating a teacher override interface. Broader language support would be a separate extension requiring new data and validation. The present component remains English-only. Wider deployment should follow evidence of reliable operation, understandable feedback and an appropriate data-governance process.

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GLOSSARY

Glossary of specialised terms

Term	Meaning in this thesis
Accepted note	The text approved by the verification and reconstruction policy for use in synthesis
Atomic claim	A factual statement whose meaning can be assessed independently while retaining necessary context
Calibration	Agreement between predicted confidence and observed correctness on a defined evaluation
Citation adequacy	Whether a cited passage supports the complete decision, beyond merely having a valid identifier
Conservative reconstruction	Rebuilding a note using supported claims while excluding unresolved or rejected claims
Decision coverage	The proportion of processed claims for which an educational verdict can be made
Evidence gate	A rule requiring valid selected evidence before accepting a content verdict
Fail-closed behaviour	Withholding a decision when a required condition or permission is absent
Provenance	The recorded origin and transformation history of text, claims, evidence and generated outputs
Re-verification	A fresh check of a proposed correction before it enters the accepted note
Semantic support	A meaningful relationship in which evidence justifies the complete claim or verdict
Synthesis drift	Introduction or alteration of meaning while converting accepted text into another format

APPENDICES

Appendix A Corpus and Run Inventory

Tables A.1 and A.2 define the prospective corpus inventory and run manifest. Complete and version these records when evaluation is executed; no completed collection or run is implied.

Table A.1: Curriculum source inventory fields

Field	Value to record
Document identifier	Stable internal identifier
Source title and grade	Exact title and Grade 10 or Grade 11
Edition and source location	Year, publisher or issuing body, and authorised location
Language	English
Permission and access	Basis for use and restrictions on redistribution
Page mapping	PDF page index and printed page number where different
Extraction version	Tool, settings, date and known missing regions
Chunking configuration	Length, overlap and treatment of tables and headings
Index identity	Embedding model, document hash and index version

Table A.2: Evaluation run manifest

Field	Value to record at execution
Run identifier and date	Unique identifier and timestamp
Local verification model	Llama 3.2 3B; record exact deployed tag and digest
Cloud OCR and synthesis model	Gemma 4 Cloud; record endpoint and available revision
Generation configuration	Temperature, limits and relevant sampling settings
Retrieval configuration	Embeddings, corpus version, candidate limits and fusion settings
Reranker state	Enabled or disabled; model revision if enabled
Machine and network	CPU, GPU if used, RAM, operating system and connection conditions
Dataset partition	Development, validation or held-out test
Prompt and code revision	Version identifiers for the executed configuration
Failures and retries	Counts, causes and retry policy

Appendix B Teacher Annotation Instrument

Reviewers should first read the approved curriculum context and the label definitions. They should annotate independently before discussing disagreements. A claim must be assessed in its note context, while the evidence record must identify the exact source location used for the judgement. Do not use the system's predicted label as the reference answer. Table B.1 defines the claim record.

For each note, record the anonymous note ID, grade, science unit, input origin and capture-quality category. Produce a reference transcription that preserves visible wording and identifies genuinely unreadable regions. Record scientific notation separately where ordinary text representation loses information. Do not correct scientific errors while creating the OCR reference.

Table B.1: Claim annotation record

Field	Reviewer entry
Claim ID and source fragment	Exact fragment and location in the note
Reference claim	Self-contained wording preserving the student's assertion
Reference verdict	Correct, incorrect, incomplete or insufficient evidence
Evidence	Document ID, page and supporting or contradicting passage
Reason	Concise justification tied to the evidence
Correction if needed	Minimum sufficient replacement supported by the source
Ambiguity	Missing context, unclear notation or multiple reasonable interpretations
Adjudication	Initial labels, final decision and explanation of disagreement

Use correct when the available evidence supports the complete assertion in its intended context. Use incorrect when it contradicts the assertion. Use incomplete when an essential condition or part of the explanation is missing under the agreed rubric. Use insufficient evidence when the permitted sources cannot settle the claim. Verification failed is a system state and should not be assigned as a scientific reference label.

For generated resources, rate correctness, source faithfulness, clarity and useful coverage separately. Record an example for each problem. Even a scientifically true addition violates the source restriction when it is unsupported by the accepted note.

Appendix C Software Test Catalogue

Table C.1 defines expected test behaviour. They are not records of executed tests. All actual outputs and statuses must be completed from the deployed system, with a run manifest linked to each execution. A screenshot can illustrate an output, but the underlying response or trace should be retained where possible.

Table C.1: Proposed functional and failure tests

ID	Input or condition	Expected behaviour
TC01	Valid English note image	Editable transcription or explicit OCR error
TC02	Oversized or unsupported upload	Clear rejection before model processing
TC03	Unclear scientific symbol	Uncertainty retained for user review
TC04	Empty reviewed transcript	No fabricated claim or score
TC05	Mixed correct and incorrect claims	Separate claim records and cited decisions
TC06	Invented evidence identifier	Verdict withheld or safely downgraded
TC07	No relevant curriculum passage	No automatic false label
TC08	Empty approved-domain list	No web-based decision
TC09	Timeout on one claim	Failed claim isolated; other results retained
TC10	No decidable claims	Unavailable score; explicit state counts
TC11	Unsupported proposed repair	Replacement excluded from accepted note
TC12	Edit after verification	Earlier results marked stale or rechecked
TC13	Synthesis with no accepted content	Clear empty-state handling
TC14	Mind-map edge to missing node	Invalid edge removed or output rejected
TC15	Missing video captions	Explicit unavailability, not invented summary
TC16	Cloud synthesis unavailable	Existing local verification result preserved
TC17	Missing FAISS index	Grounded verification suspended with a clear error
TC18	Note exceeds claim limit	Processing boundary or continuation made visible

For each test, record the executor, execution date, configuration ID, actual result, pass/fail status and evidence location. A test passes only when the observed response satisfies the expected condition. If a requirement changes, revise the test definition before comparing later runs. Unexecuted tests remain unexecuted; they must not be counted as successful.

Appendix D Results Capture and Reporting Structure

Table D.1 defines how future results should be captured. It contains no experimental observations. The final results release should retain note-level records so that a reviewer can reproduce aggregate metrics and identify excluded or failed cases.

Table D.1: Required result fields by analysis

Analysis	Required fields
OCR	Note ID, reference length, substitutions, deletions, insertions and notation errors
Extraction	Reference claim count, matched predictions, missed claims and duplicates
Retrieval	Query ID, gold evidence IDs, retrieved ranks and corpus version
Verification	Reference label, predicted label, citation adequacy and failure state
Reconstruction	Valid source claims, retained claims, accepted repairs and unsupported additions
Synthesis	Output type, reviewer scores, unsupported statements and coverage
Operation	Per-stage duration, total duration, retries, provider errors and usage

Report the number of notes, images, extracted claims and reference claims separately. Include the number of reviewers and the proportion independently double-annotated. Present the confusion matrix with labelled rows and columns, and state how failed requests were handled. Where a metric is undefined, report it as unavailable rather than converting it to zero without explanation.

The principal comparison table should show each baseline on the same held-out data, with uncertainty intervals where feasible. A second table should show the operational cost and failure rate. Representative examples should include at least one successful correction, one appropriate abstention, one retrieval failure and one synthesis error if those cases occur. Select examples to explain mechanisms, not only to make the system look successful.

Appendix E Proposed Usability Session

The usability session is intended to follow institutional approval and consent. It should use non-sensitive notes and make clear that the system does not assign examination marks. The facilitator should avoid explaining the interface during a task unless help is requested; otherwise, task completion may reflect coaching rather than usability.

Task 1 asks the participant to upload a note and correct one recognition error. Task 2 asks the participant to find the evidence for a verdict. Task 3 asks the participant to explain the difference between an incorrect claim and insufficient evidence. Task 4 asks the participant to compare an original sentence with its accepted correction. Task 5 asks the participant to inspect a flashcard against the accepted note. Task 6 asks the participant to recover from a deliberately unavailable synthesis service.

Record completion, time, requests for help, mistaken interpretations and the recovery path. Ask which part of the feedback was unclear and whether the participant knew what to do next. Preference questions should follow the tasks, not replace observation. Any claim about learning benefit requires a separate study with learning measures; this usability protocol alone cannot establish improved retention or examination performance.

Appendix F Academic Integrity and AI Use Disclosure

Generative AI assistance was used in preparing this thesis draft for organisation, language drafting, literature discovery, technical diagram preparation and document formatting. The research description was developed from the project materials and the author's clarifications of the component scope and model responsibilities. AI assistance was not used to create experimental observations, teacher judgements, participant records or performance measurements.

The drafting workflow included checking selected references against primary publication pages and official documentation, consolidating repeated sources and distinguishing design claims from unmeasured outcomes. This assistance does not transfer responsibility for the thesis from the candidate. Before submission, the candidate must review the technical descriptions against the actual implementation, verify every citation, complete the outstanding empirical evaluation and follow the institution's disclosure requirements.

Representative instructions were to follow the prescribed individual-thesis structure and formatting, restrict the component to English, distinguish cloud OCR and synthesis from local verification, retain unresolved claims outside the accepted note, and avoid reporting unexecuted tests as passed. No claim is made that the resulting prose will obtain a particular similarity or automated-detection score.